



Nasaq

The Intelligent School Timetable Generator

Technical Report · Working Prototype · Kalba Waterfront Competition

A complete school timetable in seconds... with zero clashes



Generates



Prevents clashes



Covers absences



Protects names



Saves energy

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00 Executive Summary

Nasaq is a software application that automatically generates a complete school timetable using a search-and-optimisation AI algorithm. The school enters teachers, subjects, unavailable times and priorities; Nasaq explores hundreds of thousands of arrangements and selects one that satisfies every hard constraint while optimising educational and environmental preferences. It then guards the timetable against invalid edits, covers teacher absences in one click, and protects names with three privacy levels.



Measured on the demo dataset (8 classes · 13 teachers · 240 weekly lessons); values vary slightly per run.

Why it matters – Because the timetable touches every teacher, student and family every day. Nasaq turns a long, error-prone manual task into a precise, explainable process aligned with Zero Bureaucracy, the National AI Strategy, sustainability and the Year of Community.

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01 The Problem

In a mid-sized school, 240 weekly lessons are distributed across 8 classes, 13 teachers and 3 shared rooms (science lab, computer lab, gym). Each teacher serves several classes, some are unavailable at certain times, and rooms have capacity limits. A single mistake, such as a teacher in two classes at once, breaks the whole timetable.

- Manual arrangement takes days of trial and error (estimate).
- Midday PE exposes students to Kalba's intense heat.
- Scattered lab lessons keep air-conditioning running all day.
- Sudden absences mean morning phone calls and paperwork with no clear fairness rule.

02 Solution Overview



Generator screen: the timetable cost falls during the search down to zero clashes.

Module	What it does
Smart generator	Generates and optimises the full school timetable in seconds.
Class & teacher views	Clear views with safe swaps and explained clash rejection.
Absence & cover	Ranks substitutes with a public formula, one-click cover, ready message.
Rooms & energy	Occupancy map and estimated AC-hour savings.
Constraints & data	Unavailability, preference weights and teacher names.
Digital trust	3 privacy levels, 4 roles and a live access log.
Ask Nasaq	Arabic questions answered from the live timetable.

03 The Engine in Detail

3.1 Representation

A week = 5 days × 6 periods = 30 slots per class. Each class holds exactly its weekly lesson count per subject (total 30). The teacher is derived from (class, subject) and the special room from the subject. Every class is therefore always full, and the search becomes a re-ordering of each class's lessons.

3.2 Cost function

```
cost = 1000 × HARD + SOFT
HARD = ∑ teacher double-bookings + ∑ room over-capacity + ∑ unavailable slots + ∑ same-
subject repeats per day
SOFT = 3·(math/science after period 4) + 4·(PE after period 4) + 2·(teacher idle gaps)
      + 5·(teacher load > 5/day) + 3·(run > 4 consecutive) + 3·(idle periods inside a
room's daily span)
```

Soft weights are adjustable in the Constraints tab; the values above are defaults. The cost decomposes by day, since every constraint depends only on slots within the same day.

3.3 Simulated annealing

- Start from a random arrangement per class.
- Move: pick a random class and swap two of its lessons.
- Incremental evaluation: only the two affected days are re-scored, about 2.5x cheaper than re-scoring the whole week.
- Acceptance: improvements always; worse moves with probability $e^{(-\Delta/T)}$ to escape local minima.
- Cooling: 400,000 moves with geometric temperature decay from 40 to 0.04.
- Re-heating: if a hard clash remains, another round runs at $T_0 = 8$ (up to 3 rounds). The best solution is always kept.
- Performance: the search runs in chunks so the UI stays smooth while the class grid and cost curve update live.

Why is this AI? School timetabling is a computationally hard (NP-hard) constraint-satisfaction problem. Guided search and approximate optimisation are a core branch of AI, used in university and industrial scheduling systems. Nasaq does not use a language model, and we say so plainly.

04 Hard and Soft Constraints

Type	Constraint	Rationale	Weight
Hard	No teacher in two classes at once	Physically impossible	1000
Hard	Room capacity: science lab 2, computer lab 1, gym 1	Space limits	1000
Hard	Teacher unavailable times	Teacher commitments	1000
Hard	No subject repeated in a day for a class	Spaced learning	1000
Soft	Math & science in periods 1-4	Higher morning focus	3
Soft	PE before 10:50	Kalba heat and student safety	4
Soft	Fewer teacher idle gaps	Teacher wellbeing	2
Soft	At most 5 lessons per teacher per day	Prevent overload	5
Soft	No more than 4 consecutive	Rest breaks	3
Soft	Cluster special-room lessons	Fewer AC hours	3

Demo dataset

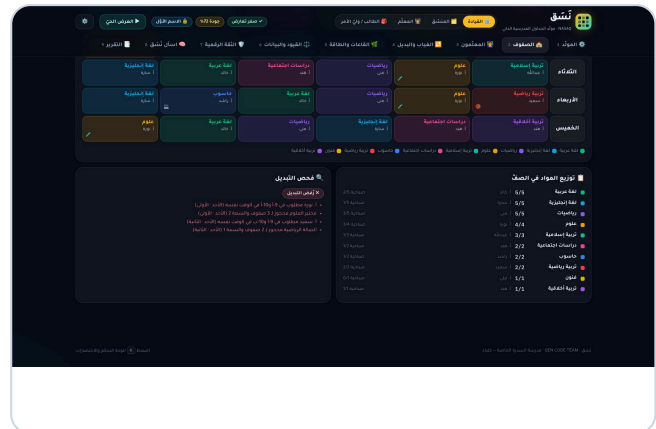
8 classes (grades 7-10, A/B) · 10 subjects · 13 teachers with fictitious names · loads from 8 to 24 lessons · realistic unavailability (an art teacher available only three days, a teacher unavailable for the first two Sunday periods, ...).

Subject	Lessons/week	Special room
Arabic	5	–
English	5	–
Mathematics	5	–
Science	4	Science lab
Islamic education	3	–
Social studies	2	–
Computing	2	Computer lab
Physical education	2	Gym
Art	1	–
Moral education	1	–

05 Error Protection and Safe Swaps

A coordinator may swap any two lessons in the same class. Nasaq scores the timetable before and after; if hard clashes increase, the swap is refused with a plain-language reason, e.g. “Ms Noura required in 9-A and 10-A at the same time (Sunday · period 1)”. Safe target cells are outlined green and risky ones red at selection time.

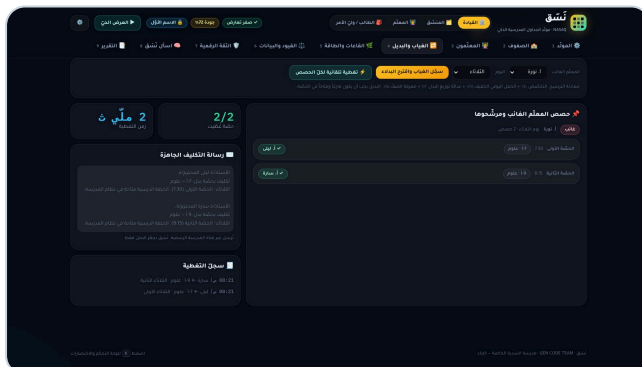
“Suggest safe swaps” checks every lesson pair in the class (435 pairs) and proposes the best improvements with zero clashes.



06 Smart Absence Coverage

When a teacher is marked absent for a day, Nasaq lists each lesson and ranks the substitutes who are free and available at that time with a public 100-point formula:

$$\text{score} = 40 \cdot [\text{same subject}] + 25 \cdot (1 - \text{dayLoad}/5) + 20 \cdot (1 - \text{subs}/\text{maxSubs}) + 15 \cdot [\text{already teaches this class}]$$



- “Auto-cover” assigns the top candidate per lesson in milliseconds.
- The fairness term down-ranks teachers who have covered often.
- A ready assignment message per substitute and a timestamped cover log.
- Cover lessons appear in the substitute’s timetable instantly.

07 Rooms and Energy

In typical operation, special-room air-conditioning runs for the whole school day. Nasaq clusters each room's lessons into contiguous blocks, so AC can run only from the first to the last lesson. The room-clustering term in the cost function is what drives the algorithm toward this.



$$\text{saved_kWh/week} = \sum_{\text{rooms}} \sum_{\text{days}} (6 - \text{span_periods}) \times 0.75 \text{ h} \times \text{room_kW}$$

room_kW: science lab 9 · computer lab 7 · gym 14 tariff: 0.38 AED/kWh 36 school weeks

≈272

kWh / week

≈9,800

kWh / year

≈3,700

AED / year

≈3,900

kg CO₂ / year

Transparency note: These are estimates from stated assumptions, not metered readings.

The carbon factor used is 0.4 kg CO₂/kWh. In a real deployment the assumptions are replaced with the school's bills and equipment ratings.

A health dimension too: the PE-before-10:50 rule protects students from Kalba's heat, a clear local priority.

08 Privacy and Digital Trust

Level	Example	Use
 Full name	Ms Noura Al Kaabi	Administration, in office
 First name (default)	Ms Noura	Public display
 Codes only	07-م	Competitions and exposed screens

The level applies to every screen, dropdown, assignment message, cover log, Ask-Nasaq answer, report and exported Excel file. The student/parent role never sees full names even if the level is “full”. Real names are entered in the Constraints & Data tab and stored only on the device.

Permission matrix

Resource	Leadership	Coordinator	Teacher	Student/Parent
Generate timetable	✓	✓	✗	✗
Swap lessons	✓	✓	✗	✗
All class timetables	✓	✓		
All teacher timetables	✓	✓	 own + availability	✗
Absence & cover	✓	✓	✗	✗
Constraints	✓	✓	✗	✗
Full names	✓		✗	✗
Reports & export	✓		✗	✗
Privacy level	✓	✗	✗	✗
Access log	✓	✗	✗	✗

Permissions are enforced, not cosmetic: tabs appear per role, swaps are blocked in view-only mode, and Ask-Nasaq in the student role answers class-timetable questions only. Every action is written to a live, exportable access log (up to 120 entries). No data is sent to any server or third party.

09 Testing and Results

We tested the prototype automatically in Chromium (Playwright) on the final build itself:

Test	Method	Result
Generation correctness	6 consecutive generations	0 hard clashes every time
Generation time	Measured with performance.now()	1.3 - 1.6 s
Optimisation quality	Penalty before/after	≈260-350 → ≈93-119
Name leakage	4 roles x all tabs x “first” & “codes” levels	0 family names visible
Exported files	CSV and TXT inspection	Respect privacy level
Permissions	Swap & query as student	Blocked with clear message
Persistence	Page reload	Settings & timetable kept
Live demo	Full 10-step run	No runtime errors

The screenshot displays the 'الثقة الرقمية والخصوصية' (Digital Trust and Privacy) section of the Nasaq application. The interface is in Arabic and includes a top navigation bar with various icons and a main content area with several panels.

الثقة الرقمية والخصوصية

من يرى ماذا؟ صلاحيات واضحة لكل دور وأسماء مخفية، وسجل وصول حتى لكل إجراء. كل ذلك يعمل على هذا الجهاز فقط ولا يُرسل لأي جهة.

ملخص الحماية

- إجراء مسجل: 11
- اسم الأول: مستوى الخصوصية
- أدوار صلاحيات: 4
- بيانات تُرسل للخارج: 0

مستوى خصوصية الأسماء

اسم الكامل، اسم الأول، رمز فقط

المستوى: الاسم الكامل، الاسم الأول، رمز فقط

مثال: أ. نورة الكعبي

الاستخدام: الإدارة داخل المكتب، الأعراسي - العرض العام، للمسابقات والاشعارات المخشوفة

رموز فقط: 07-م

يطلق المستوى على كل الاشعارات ورسائل التكيف واجابات اسائل تسقيع والتقارير والمفاتيح المصارة. دور الطالب وولي الامر لا يرى الأسماء الكاملة أبداً.

مصفوفة الصلاحيات

المورد: القيادة، المنسق، المعلم، الطالب / ولي الامر

توليد الجدول: ✓

تدبير الخصص: ✓

جداول كل الصفوف: ✓

جداول كل المعلمين: ✓

الغياب والبديل: ✓

القبود وأوقات التوفر: ✓

Digital trust centre: privacy level, protection summary and access log.

10 National Alignment

Priority	How Nasaq serves it
Zero Government Bureaucracy	Timetable and cover without paper, calls or coordination meetings; ready assignment messages.
UAE National AI Strategy 2031	A search-and-optimisation algorithm that solves a hard scheduling problem and explains its decisions.
Sustainability & Net Zero 2050	Fewer special-room AC hours; PE scheduled before the heat.
Year of Community & Year of the Family	Fair load and cover distribution; a stable timetable families can rely on.
Education quality	Math and science in the most focused periods.

11 Limits and Next Steps

Candidly

- Data and names are fictitious; no real person's data.
- Energy figures are estimates with stated assumptions.
- Ask-Nasaq is a local rule engine, not a language model.
- Data lives in the device browser; no cross-device sync yet.

Next steps

- One-click import of school data from Excel.
- A pilot with school leadership under formal approval.
- Double periods, multiple buildings and shifts.
- Share class timetables with families via a secure link.

Technology – One self-contained HTML file (vanilla JavaScript, no libraries or server), embedded fonts; runs offline with a double-click in any modern browser.

GEN CODE TEAM · Al Sidra Private School – Kalba, Sharjah